spectra with infrared light, helping us to see galaxies so far away that their light has been stretched out of the visible part of the spectrum altogether.

Even cosmic microwave background observatories are getting in on the dark energy game. We saw in Chapter 2 how studying the CMB can tell us about the early universe and the origins of cosmic structure. At the time the CMB light was emitted, dark energy was completely unimportant in the universe, its effects totally swamped by the extreme densities of matter and radiation. So it may be surprising that CMB observations give us any insights into how dark energy is acting today. The trick is that all the cosmic structure we want to study—every galaxy and cluster of galaxies—is *between* us and the CMB, and each one of those objects distorts the space it's in just a little bit with its gravity.

Imagine you have a snapshot looking down into a clear-water pond at the pebbles below. Even if you don't know exactly where every pebble should have been placed, or all their exact shapes, you could probably tell the difference between very still water and water that has some ripples in it by noticing distortions in how the pebbles look, because you have a sense of what pebbles should look like in general. In a similar way, we understand the cosmic microwave background so well that we can see, at least in a statistical sense, the tiny distortions in its light due to all the stuff between here and there. This is called CMB lensing, and it's a fantastic tool for studying the growth of cosmic structure. New CMB observatories will help us refine the method, but we've already used CMB lensing to make a map of ALL OF THE DARK MATTER IN THE OBSERVABLE UNIVERSE. Granted, the map is an extremely low-resolution, blurry kind of map, like a map of the world reproduced from memory with fingerpaints, but still, it's pretty impressive that this is a thing we can do at all.

Renée Hložek, a cosmologist at the University of Toronto, uses the CMB and galaxy surveys to better understand our cosmological model, with a particular interest in dark energy and the universe's ultimate fate. She points out that combining data between things like VRO and new CMB observatories will become especially powerful as each data set improves. Using a technique called cross-correlation, we can take what we know about the positions of individual objects from galaxy catalogs and compare that with what we know about the largest-scale distribution of matter from CMB lensing. This can give us more precise results that make it harder to miss any deviations from the Concordance Model. Alternative theories that